

R Code and Output for Chao estimate other subfamilies

Load packages

```
library(dplyr)
library(SpadeR)
library(stringr)
estimator <- "    iChao1 (Chiu et al. 2014)"
```

List subfamilies from excel file

```
sub_families_list <- list.files("../Data/BIN_abundance_subfamilies/",
                                full.names = TRUE)
```

Calculate iChao1 for both reared and core and compile for all species

```
set.seed(100)
sub_family_outputs_all <- data.frame()

for(fm in sub_families_list){

  ## load data

  data_all <- read.csv(fm)

  ## filter for core and reared

  core <- select(data_all, BIN, count = Core_Frequency) %>%
    filter(!is.na(count)) %>%
    mutate(source = 'core')
```

```

reared <- select(data_all, BIN= BIN, count = Reared_Frequency) %>%
  filter(!is.na(count)) %>%
  mutate(source = 'reared')

## summarise total bins

total <-
  bind_rows(core, reared) %>%
  group_by(BIN) %>%
  summarise(count = sum(count))

## calculate shared, reared only and malaise only

shared <- unique(c(core$BIN[core$BIN %in% reared$BIN],
  reared$BIN[reared$BIN %in% core$BIN]))

reared_only <- reared$BIN[!reared$BIN %in% core$BIN]

core_only <- core$BIN[!core$BIN %in% reared$BIN]

## Estimate iChao for core -- use try catch for subfamilies with too few bins

core_estimate <- tryCatch({
  ChaoSpecies(core$count, "abundance", k=10, conf=0.95)},
  error = function(e) {
    template_result <- ChaoSpecies(reared$count, "abundance", k=10, conf=0.95)
    na_matrix <- template_result
    na_matrix$Species_table[] <- NA
    na_matrix$Basic_data_information[] <- NA
    return(na_matrix)
  })

## Estimate iChao for reared

reared_estimate <- ChaoSpecies(reared$count, "abundance", k=10, conf=0.95)

subfamily <- str_remove(str_remove(fm, "../Data/BIN_abundance_subfamilies//"), ".csv")

## compile results in data frame

sub_family_outputs_1 <- data.frame(subfamily = subfamily,
  total_n_bin = nrow(total),

```

```

shared_bins = length(shared),
proportion_shared = length(shared)/nrow(total),
nreared_total_bin = nrow(reared),
reared_only = length(reared_only),
core_only = length(core_only),
reared_Chao1_estimate = reared_estimate$Species_table[estimator,
reared_Chao1_se = reared_estimate$Species_table[estimator,
reared_Chao1_95_lower = reared_estimate$Species_table[estimator,
reared_Chao1_95_upper = reared_estimate$Species_table[estimator,
reared_Chao1_CV = reared_estimate$Basic_data_information[estimator,
reared_Chao1_Coverage = reared_estimate$Basic_data_information[estimator,
core_total_bin = nrow(core), core_Chao1_estimate = core_estimate$Species_table[estimator,
core_Chao1_se = core_estimate$Species_table[estimator,
core_Chao1_95_lower = core_estimate$Species_table[estimator,
core_Chao1_95_upper = core_estimate$Species_table[estimator,
core_Chao1_CV = core_estimate$Basic_data_information[estimator,
core_Chao1_Coverage = core_estimate$Basic_data_information[estimator,

sub_family_outputs_all <- bind_rows(sub_family_outputs_all, sub_family_outputs_1)
}

```

Warning: In this case, it can't estimate the variance of 2nd-order-jackknife estimation

Warning: In this case, it can't estimate the variance of 2nd-order-jackknife estimation

Warning: In this case, it can't estimate the variance of Homogeneous estimation

```

rownames(sub_family_outputs_all) <- NULL

sub_family_outputs_all <- sub_family_outputs_all %>%
  filter(!is.na(core_Chao1_estimate), subfamily != "Homolobinae")

```

Reared summary table

```

reared_summary <-sub_family_outputs_all %>% select(subfamily, total_n_bin, nreared_total_bin)
mutate(reared_rank = 1:n()) %>%
mutate(reared_Chao1_estimate = round(reared_Chao1_estimate)) %>%
mutate(reared_completeness = round(nreared_total_bin/reared_Chao1_estimate,2)) %>%
arrange(desc(reared_completeness)) %>%
mutate(reared_completeness_rank = 1:n()) %>%
arrange(desc(total_n_bin))

```

```
reared_summary
```

	subfamily	total_n_bin	nreared_total_bin	reared_Chao1_estimate
1	Microgastrinae	1168	889	1131
2	Campopleginae	333	243	298
3	Cheloninae	190	77	124
4	Euphorinae	186	58	78
5	Mesochorinae	184	151	218
6	Rogadinae	168	98	157
7	Agathidinae	150	141	168
8	Ophioninae	108	101	131
9	Metopiinae	88	63	87
10	Macrocentrinae	57	41	117
11	Orgilinae	55	28	32
12	Anomaloninae	50	44	59
13	Tryphoninae	31	12	18

	reared_rank	reared_completeness	reared_completeness_rank
1	1	0.79	4
2	2	0.82	3
3	7	0.62	12
4	9	0.74	7
5	3	0.69	9
6	6	0.62	11
7	4	0.84	2
8	5	0.77	5
9	8	0.72	8
10	11	0.35	13
11	12	0.88	1
12	10	0.75	6
13	13	0.67	10

```
reared_summary$nreared_total_bin %>% mean() %>% round()
```

```
[1] 150
```

```
reared_summary$reared_Chao1_estimate %>% sd() %>% round()
```

```
[1] 290
```

Core summary table

```
core_summary <-sub_family_outputs_all %>% select(subfamily, total_n_bin, core_total_bin, co
mutate(core_rank = 1:n()) %>%
mutate(core_Chao1_estimate = round(core_Chao1_estimate)) %>%
mutate(core_completeness = round(core_total_bin/core_Chao1_estimate,2)) %>%
arrange(desc(core_completeness)) %>%
mutate(core_completeness_rank = 1:n()) %>%
arrange(desc(total_n_bin))

core_summary
```

	subfamily	total_n_bin	core_total_bin	core_Chao1_estimate	core_rank
1	Microgastrinae	1168	388	725	1
2	Campopleginae	333	161	270	2
3	Cheloninae	190	117	278	4
4	Euphorinae	186	136	213	3
5	Mesochorinae	184	48	83	6
6	Rogadinae	168	79	122	5
7	Agathidinae	150	32	46	7
8	Ophioninae	108	23	42	11
9	Metopiinae	88	30	82	8
10	Macrocentrinae	57	26	57	10
11	Orgilinae	55	30	51	9
12	Anomaloninae	50	10	18	13
13	Tryphoninae	31	21	63	12
	core_completeness	core_completeness_rank			
1	0.54	9			
2	0.60	4			
3	0.42	11			
4	0.64	3			
5	0.58	6			
6	0.65	2			
7	0.70	1			
8	0.55	8			
9	0.37	12			
10	0.46	10			
11	0.59	5			
12	0.56	7			
13	0.33	13			

```
core_summary$core_Chao1_estimate %>% sd() %>% round()
```

```
[1] 192
```

```
core_summary$core_completeness %>% sd() %>% round(2)
```

```
[1] 0.11
```

Combined reared and core completeness

```
all_summary <- select(reared_summary, subfamily, total_n_bin, reared_completeness) %>%
  left_join(select(core_summary, subfamily, total_n_bin, core_completeness)) %>% mutate(mean_
  arrange(desc(mean_completeness)) %>%
  mutate(completeness_rank = 1:n()) %>%
  arrange(desc(total_n_bin))
```

Joining with `by = join\_by(subfamily, total\_n\_bin)`

```
all_summary
```

	subfamily	total_n_bin	reared_completeness	core_completeness
1	Microgastrinae	1168	0.79	0.54
2	Campopleginae	333	0.82	0.60
3	Cheloninae	190	0.62	0.42
4	Euphorinae	186	0.74	0.64
5	Mesochorinae	184	0.69	0.58
6	Rogadinae	168	0.62	0.65
7	Agathidinae	150	0.84	0.70
8	Ophioninae	108	0.77	0.55
9	Metopiinae	88	0.72	0.37
10	Macrocentrinae	57	0.35	0.46
11	Orgilinae	55	0.88	0.59
12	Anomaloninae	50	0.75	0.56
13	Tryphoninae	31	0.67	0.33

	mean_completeness	completeness_rank
1	0.66	5
2	0.71	3
3	0.52	11
4	0.69	4

5	0.64	8
6	0.64	9
7	0.77	1
8	0.66	6
9	0.54	10
10	0.41	13
11	0.74	2
12	0.66	7
13	0.50	12

```
all_summary$mean_completeness %>% sd() %>% round(3)
```

```
[1] 0.104
```

```
shared_summary <- sub_family_outputs_all %>% select(subfamily, reared_only, core_only, shared_only) %>%
  mutate(proportion_shared = round(proportion_shared, 2)) %>%
  arrange(desc(proportion_shared)) %>%
  mutate(shared_rank = 1:n()) %>%
  arrange(desc(total_n_bin))

shared_summary
```

	subfamily	reared_only	core_only	shared_bins	total_n_bin
1	Microgastrinae	780	279	109	1168
2	Campopleginae	172	90	71	333
3	Cheloninae	73	113	4	190
4	Euphorinae	50	128	8	186
5	Mesochorinae	136	33	15	184
6	Rogadinae	89	70	9	168
7	Agathidinae	118	9	23	150
8	Ophioninae	85	7	16	108
9	Metopiinae	58	25	5	88
10	Macrocentrinae	31	16	10	57
11	Orgilinae	25	27	3	55
12	Anomaloninae	40	6	4	50
13	Tryphoninae	10	19	2	31

	proportion_shared	shared_rank
1	0.09	5
2	0.21	1
3	0.02	13
4	0.04	12

5	0.08	7
6	0.05	11
7	0.15	3
8	0.15	4
9	0.06	8
10	0.18	2
11	0.05	10
12	0.08	6
13	0.06	9

```
shared_summary$proportion_shared %>% sd() %>% round(2)
```

```
[1] 0.06
```

```
shared_summary$total_n_bin %>% sum()
```

```
[1] 2768
```